Yongjung Kim | CV

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Positions

Assistant Professor — **Seoul, Korea**
Sejong University — *2025.03 – present*
School of Liberal Studies; Dept. of Physics and Astronomy

Senior Researcher — **Daejeon, Korea**
Korea Astronomy and Space Science Institute (KASI) — *2023.07 – 2025.02*
Space Astronomy Group (PI: Dr. Woong-Seob Jeong)

Sejong Science Fellow — **Daegu, Korea**
Kyungpook National University — *2021.09 – 2023.06*
Advisor: Prof. Minjin Kim

KIAA Fellow — **Beijing, China**
Kavli Institute for Astronomy and Astrophysics at Peking University — *2019.11 – 2021.09*
Advisor: Prof. Linhua Jiang

Postdoctoral Fellow — **Seoul, Korea**
Research Institute for Basic Sciences at Seoul National University — *2019.09 – 2019.10*
Advisor: Prof. Myungshin Im

Education

Seoul National University — **Seoul, Korea**
Ph.D. in Astronomy — *2013.03 – 2019.08*
Thesis title: Survey of Faint Quasars at High Redshifts
Supervisor: Prof. Myungshin Im

Seoul National University — **Seoul, Korea**
B.S. in Astronomy — *2009.03 – 2013.02*
Minor: Physics

Research Interests

Observational Cosmology with Quasars

- High-redshift Quasar Survey with Infrared Medium-deep Survey (IMS)
- Contribution of faint quasars to the cosmic reionization and ionizing backgrounds
- Growth of the supermassive black holes with their host galaxies at various redshifts
- Demography of quasars along the cosmic time
- Multi-wavelength surveys (participating in IMS, DESI, CSST & SPHEREx)

Research Grants

The Sejong Science Fellowship
Funded by National Research Foundation of Korea ($420,000) — *2021-2026*
Subject: Cosmic Evolution of Quasar-Galaxy by developing an Integrated Analysis Model for the Observational Big Data

The 2020 China Postdoc Science Special Grant
Funded by China Postdoctoral Science Foundation ($26,000) — *2020-2021*
Subject: Quasar and Host Galaxy Properties with the Newest Large Survey Data

The 2020 China Postdoc Science General Grant
Funded by China Postdoctoral Science Foundation ($12,000) *2020-2021*
Subject: Quasar and Host Galaxy Properties with the Newest Large Survey Data

Top 100 Fellowship
Postdoc International Exchange Program at PKU ($7,000) *2019-2021*
Subject: Enhanced Studies on High-redshift Quasars

Honors and Awards

Scholarship for Creative Academic Performance
Brain Korea 21 Program for Leading Universities and Students as a BK fellow ($ 58,000) *2013 – 2019*
awarded by National Research Foundation of Korea

Academic Excellence Scholarship
Partial tuition scholarship ($2,000) *2014, 2015*
awarded by Seoul National University

SNU in Global Research Awards
2nd place *2013*
awarded by Office of International Affairs at Seoul National University

Best Poster Presentation Awards at the 2012 Fall KAS Meeting
1st place *2012*
awarded by Korean Astronomical Society

Lotte Scholarship
Full tuition scholarship ($5,000) *2012*
awarded by Lotte Foundation

Presidential Science Scholarship
Full tuition scholarship ($10,000) *2009 – 2010*
awarded by National Research Foundation of Korea

Observational Experience

Classical/Remote Observations

BOAO 1.8 m Telescope **Bohyunsan Optical Astronomy Observatory**
Longslit Spectrograph
2023 January 27-31 (5 nights)

Palomar 200 inch Telescope **Palomar Observatory**
DBSP (*Remote*) (*Telescope Access Program*)
2 nights in 2020B, 2 nights in 2021A

Magellan Baade 6.5 m Telescope **Las Campanas Observatory**
IMACS & FIRE
2015 January 18-19, September 11-13; 2016 December 3-5; 2018 September 9-10 (10 nights)

Otto Struve 2.1 m Telescope **McDonald Observatory**
SQUEAN & CQUEAN
2014 June 3-8, November 3-9; 2015 June 19-28; 2016 July 25-28; 2017 February 1-11, April 19-26, September 16-24, December 26-31; 2018 April 16-25; 2019 February 5-14 (81 nights)

Maidanak 1.5 m Telescope **Maidanak Observatory**
SNUCAM
2013 August 2-7 (6 nights)

Observations awarded as PI

James Clerk Maxwell Telescope **East Asian Observatory**
SCUBA-2
9.00 hr in 2018A

Gemini 8 m Telescopes **Gemini Observatory**
GMOS-N, GMOS-S, & FLAMINGOS-2 (*K-GMT Science Program*)
9.00 hr in 2016B; 13.00 hr in 2017B; 7.92 hr in 2018A; 14.00 hr in 2018B; 10.14 hr in 2019A; 6.9 hr in 2020A

Atacama Large Millimeter/submillimeter Array
12m Arrays
3.6 hr in Cycle 4; 2.6 hr in Cycle 5

Observations awarded as Co-PI

Gemini 8 m Telescope **Gemini Observatory**
GMOS-S (*K-GMT Science Program*)
1 night in 2015A; 6.70 hr in 2016A; 24.00 hr in 2017A (for thesis; PI: Myungshin Im)

Academic References

Prof. Myungshin Im
Department of Physics and Astronomy, Seoul National University, Korea
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Prof. Minjin Kim
Department of Astronomy, Yonsei University, Korea
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Dr. Woong-Seob, Jeong
Korea Astronomy and Space Science Institute, Korea
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Dr. Bomee, Lee
Korea Astronomy and Space Science Institute, Korea
bmlee@kasi.re.kr

Dr. Danidel, Masters
California Institute of Technology with NASA/JPL Infrared Processing and Analysis Center, USA
dmasters@ipac.caltech.edu

Prof. Linhua Jiang
Kavli Institute for Astronomy and Astrophysics, Peking University, China
jiangKIAA@pku.edu.cn

Refereed Publications

⋆ Quick Link to ADS Library

First or Corresponding Author

12. Direct Observational Evidence that Higher-luminosity Type 1 Active Galactic Nuclei Are More Commonly Triggered by Galaxy Mergers
– Yoon, Y., **Kim, Y.**, Kim, D., et al. 2025, A&A, in press
11. Exploring Unobscured QSOs in the Southern Hemisphere with KS4
– **Kim, Y.**, Kim, M., et al. 2024, ApJS, 972, 171
10. Eddington Ratios of Dust-obscured Quasars at $z \lesssim 1$: Evidence Supporting Dust-obscured Quasars as Young Quasars
– Kim, D., **Kim, Y.**, Im, M., et al. 2024, A&A, 690, 283
9. Red Type-1 Quasars after Cosmic Noon and Impact on L_{UV}-related Quasar Statistics
– **Kim, Y.**, Kim, D., et al. 2024, ApJ, 972, 171
8. The Infrared Medium-deep Survey. IX. Discovery of Two New $z \sim 6$ Quasars and Space Density down to $M_{1450} \sim -23.5$ mag
– **Kim, Y.**, Im, M., et al. 2022, AJ, 164, 114
7. Pure Density Evolution of the Ultraviolet Quasar Luminosity Functions at $2 \lesssim z \lesssim 6$
– **Kim, Y.** & Im, M., 2021, ApJL, 910, 11
6. The Infrared Medium-deep Survey. VIII. Quasar Luminosity Function at $z \sim 5$

– **Kim, Y.**, Im, M., et al. 2020, ApJ, 904,111

5. High Star Formation Rates of Low Eddington Ratio Quasars at $z \gtrsim 6$
   – **Kim, Y.**, & Im, M. 2019, ApJ, 879, 117
4. The Infrared Medium-deep Survey. VI. Discovery of Faint Quasars at z $\sim$ 5 with a Medium-band-based Approach
   – **Kim, Y.**, Im, M., et al. 2019, ApJ, 870, 86
3. The Infrared Medium-deep Survey. IV. The Low Eddington Ratio of A Faint Quasar at z $\sim$ 6: Not Every Supermassive Black Hole is Growing Fast in the Early Universe
   – **Kim, Y.**, Im, M., et al. 2018, ApJ, 855, 138
2. Discovery of a Faint Quasar at z $\sim$ 6 and Implications for Cosmic Reionization
   – **Kim, Y.**, Im, M., et al. 2015, ApJL, 813, 35
1. Newly Discovered Footprints of Galaxy Interaction around Seyfert 2 Galaxy NGC 7743
   – **Kim, Y.**, Im, M., et al. 2015, PKAS, 30, 463

Co-author

26. The SPHEREx Satellite Mission
    – Bock, J. J., Aboobaker, A. M., Adamo, J., ..., **Kim, Y.** et al. 2025, ApJ, submitted
25. Constructing a Hydrogen Line Library for Lyα emitters at low redshifts ($z \lesssim 0.4$): Estimating dust extinction and assessing Paschen line detectability with SPHEREx
    – Song, J., Song, H., Shim, H., ..., **Kim, Y.** et al. 2025, JKAS, submitted
24. Simulating Spectral Confusion in SPHEREx Photometry and Redshifts
    – Huai, Z., Bock, J. J., Cheng, Y.-T., ..., **Kim, Y.** et al. 2025, ApJ, submitted
23. Black Hole Properties of Type-1 Active Galactic Nuclei in the North Ecliptic Pole Wide Field: I. Mid-infrared Sources with Optical Counterparts
    – Kim, D., Im, M., Shim, H., ..., **Kim, Y.** et al. 2025, ApJS, submitted
22. The Redshifts from 122 Bands: Comparative Redshift Forecast for Low-Resolution Spectra from SPHEREx and 7-Dimensional Sky Survey (7DS)
    – Bae, J., Lee, B., Im, M., ..., **Kim, Y.** et al. 2025, A&A, submitted
21. AllBRICQS: The Discovery of Luminous Quasars in the Northern Hemisphere
    – Choi, Y., Fu, Y., Im, M., ..., **Kim, Y.** et al. 2025, ApJS, 280, 73
20. The Potential of the SPHEREx Mission for Characterizing PAH 3.3 μm Emission in Nearby Galaxies
    – Zhang, E., Faisst, A. L., Crill, B., ..., **Kim, Y.** et al. 2025, ApJ, 992, 3
19. New Field OB and OBe Binaries of the SMC Wing: Observational Properties and Population Modeling
    – Vargas-Salazar, I., Oey, M. S., Eldridge, Jan J., ..., **Kim, Y.** et al. 2025, ApJ, 988, 146
18. High-cadence Optical Observations of a Normal Type Ia SN 2018kp from its Early Phase
    – Lim, G., Choi, C., Im, M., ..., **Kim, Y.** et al. 2025, JKAS, 58, 31
17. Accuracy of Stellar Mass-to-light Ratios of Nearby Galaxies in the Near-Infrared
    – Kim, T., Kim, Min, Ho, L. C., ..., **Kim, Y.** et al. 2025, AJ, 169, 44
16. Eddington Ratios of Dust-Obscured Quasars at $z \sim 2$
    – Kim, D., Im, M., Lim, G., & **Kim, Y.** 2024, JKAS, 57, 95
15. Fires in the deep: The luminosity distribution of early-time gamma-ray-burst afterglows in light of the Gamow Explorer sensitivity requirements
    – Kann, D. A., White, N. E., ..., **Kim, Y.** et al. 2024, A&A, 686, 56
14. Galaxy-Galaxy Blending in SPHEREx Survey Data
    – Kim, D., Song, H., ..., **Kim, Y.** et al. 2024, JKAS, 57, 45
13. Photometric Selection of Unobscured QSOs in the Ecliptic Poles: KMTNet in the South Field and Pan-STARRS in the North Field
    – Byun, W., Kim, M., ..., **Kim, Y.** et al. 2023, ApJS, 268, 57
12. Estimators of bolometric luminosity and black hole mass with mid-infrared continuum luminosities for dust obscured quasars: Prevalence of dust obscured SDSS quasars
    – Kim, D., Im, M., ..., **Kim, Y.** et al. 2023, ApJ, 954, 156
11. ALMA/ACA CO Survey of the IC 1459 and NGC 4636 Groups: Environmental Effects on the

Molecular Gas of Group Galaxies
– Lee, B., Wang, J., ..., **Kim, Y.** et al. 2022, ApJS, 262, 31

10. The quasar luminosity function at $z \sim 5$ via deep learning and Bayesian information criterion
– Shin, S., Im, M., & **Kim, Y.** 2022, ApJ, 937, 32
9. High-z Universe probed via Lensing by QSOs (HULQ) II. Deep GMOS Spectroscopy of a QSO Lens Candidate
– Taak, Y-. C., Im, M., **Kim, Y.**, et al. 2022, A&A, 665, 5
8. Newly Discovered $z \sim 5$ Quasars via Deep Learning and Bayesian Information Criterion
– Shin, S., Im, M., **Kim, Y.** & Jiang, L. 2022, JKAS, 55, 131
7. The Infrared Medium-deep Survey. VII. Faint Quasars at $z \sim 5$ in the ELAIS-N1 Field
– Shin, S., Im, M., **Kim, Y.**, et al. 2020, ApJ, 893, 45
6. More connected, more active: galaxy clusters and groups at $z \sim 1$ and the connection between their quiescent galaxy fractions and large-scale environments
– Lee, S.-K., Im, M., ..., and **Kim, Y.** 2019, MNRAS, 490, 135
5. Intensive Monitoring Survey of Nearby Galaxies (IMSNG)
– Im, M., Choi, C., ..., **Kim, Y.**, et al. 2019, JKAS, 52, 11
4. The Infrared Medium-deep Survey. III. Survey of Luminous Quasars at $4.7 \leq z \leq 5.4$
– Jeon, Y., Im, M., Kim, D., **Kim, Y.** et al. 2017, ApJS, 231, 16
3. Discovery of a Supercluster at $z \sim 0.91$ and Testing the ΛCDM Cosmological Model
– Kim, J.-W., Im, M., ..., **Kim, Y.** et al. 2016, ApJ, 821, 10
2. The Infrared Medium-Deep Survey. V. A New Selection Strategy for Quasars at $z > 5$ Based on Medium-Band Observations with SQUEAN
– Jeon, Y., Im, M., ..., **Kim, Y.** et al. 2016, JKAS, 49, 25
1. The Infrared Medium-Deep Survey. II. How to Trigger Radio AGNs? Hints from their Environments
– Karouzos, M., Im, M., ..., **Kim, Y.** et al. 2014, ApJ, 797, 26

Conferences

Invited Talks

- *"Hunting for Faint High-redshift Quasars with Infrared Medium-deep Survey"*, K-GMT Science Program Users Meeting 2020, On-line, 2020, November 19-20.
- *"Discoveries and Properties of High-redshift Quasars with IMS"*, Science and Evolution of Gemini Observatory 2018, San Francisco (USA), 2018, July 22-26.